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Marwadi University

INVENTION DISCLOSURE FORM FOR PATENTS

Applicant Name-Marwadi University

1. **Particulars of Inventors**

Mr./Ms/Dr.	Name (Full)	Department	Designation	Mobile No.	Email	Postal Address
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2. **Provide title of the invention:**
   EnviroWatch – AQI & Weather Monitoring Dashboard

3. **In 100 words or less, please provide an abstract or summary of the invention:**
   EnviroWatch is a full-stack monitoring system designed to provide granular, district-level air quality and weather analytics. Unlike generic apps, it utilizes the Indian National Air Quality Index (NAQI) standards and Steadman's Heat Index formula to convert raw atmospheric data into actionable health verdicts. The system features an automated background harvester that captures snapshots for 780+ districts, an interactive geospatial interface, and a predictive recommendation engine. By integrating multi-pollutant analysis with historical trend tracking, it offers a high-fidelity tool for localized environmental safety assessment and proactive health management.

4. **Detail description of the invention:(Answer to all below are required in detail)**
   a. **Problem the invention is solving**
      Standard environmental monitoring tools often suffer from two main flaws: "Data Generalization" and "Inconsistent Calibration." Most applications provide city-level data that fails to account for micro-climates within vast Indian districts. Furthermore, they often use global AQI scales (like the US-EPA standard) which don't accurately reflect the pollution thresholds established by the Central Pollution Control Board (CPCB) for the Indian subcontinent. Users are left with numbers but no clear understanding of immediate health risks like the Heat Index or specific pollutant dangers (PM2.5 vs PM10).

b. General Utility/application of the invention

EnviroWatch serves as a crucial infrastructure tool for several sectors:

- Smart City Planning: Municipalities can track pollution hotspots in real-time.
- Healthcare & Public Health: Hospitals can correlate respiratory ailment spikes with localized AQI trends.
- Individual Safety: School administrators and outdoor workers can use the "Safety Verdict" and "Heat Index" logic to decide on activity levels.
- Data Research: Its historical snapshotting capabilities provide a rich dataset for environmental studies and policy-making.

c. Advantages of the invention disclosing about the increased efficiency/efficacy

The primary advantage lies in its Automated Data Lifecycle. By utilizing a specialized BackgroundService (Hosted Service in .NET 8), the system removes manual data entry and ensures 24/7 data consistency across all 780+ districts with zero downtime.

- Efficacy: The use of the Indian NAQI breakpoint table ensures that the results are legally and scientifically relevant within India.
- Efficiency: The system uses a "Dual-Caching Strategy" via its PWA (Progressive Web App) manifest, allowing users to access the latest captured data even with spotty internet connectivity, significantly increasing its reliability in rural areas.

d. Best way of using the invention as well as possible variants

The Primary Use Case is as a public-facing dashboard where users subscribe to threshold-based alerts (e.g., getting an email when PM2.5 in their district exceeds 200 $\mu g/m^3$). Possible Variants include:

- IoT Integration: Connecting the backend to physical low-cost sensors for hyper-local street-level monitoring.
- Enterprise API: Licensing the "Safety Verdict" engine as a REST API for insurance or travel companies.
- Mobile Native: Refactoring the current PWA into a Flutter/React Native app for deeper hardware integration like push notifications.

e. Working of invention along with Drawing, schematics and flow diagrams if required with complete explanations

The system operates on an Isomorphic MVC Architecture. The backend (ASP.NET Core) features a DataSnapshotService that runs a cron-like cycle every hour, fetching JSON data from weather satellites (OpenWeatherMap) and pollution sensors.

- The Core Engine: It parses these JSON packets and runs a "Sub-Index Calculation" for six major pollutants (PM2.5, PM10, O3, NO2, SO2, CO). The highest sub-index becomes the final AQI.
- The Visualizer: This data is then projected onto a Leaflet.js-based geospatial map where districts are color-coded based on severity.
- The Logic Layer: Simultaneously, the RecommendationService applies the Heat Index formula (Temperature + Humidity) and CPCB health guidelines to generate a simplified "Safety Verdict" (Safe, Caution, or Stay Indoors).

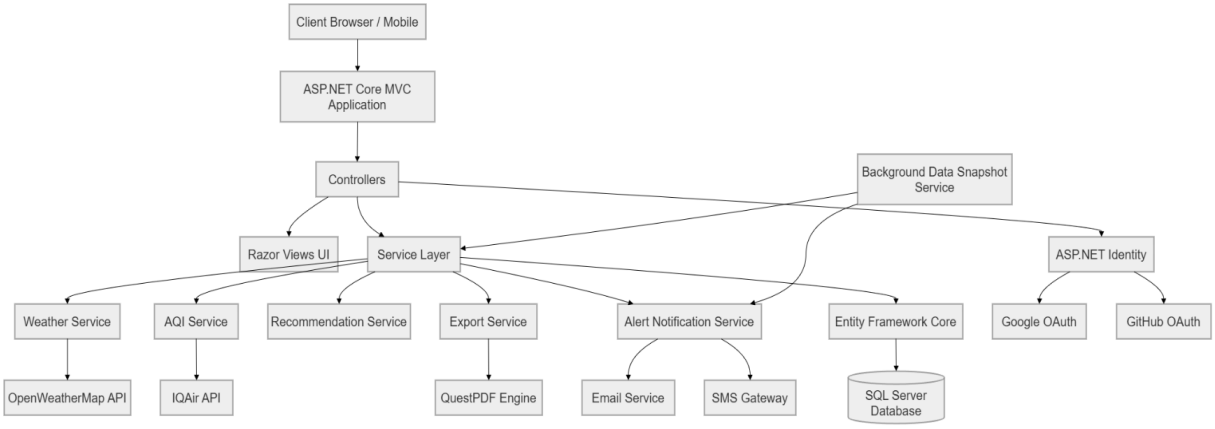


Figure - System Architecture

5. **Have you conducted Primary Patent Search? Yes / No (if yes, attach the patent search results)**
   No

6. **Existing state-of-the-art and prior arts: (Brief background of the existing knowledge/product/process in the market)**
   The current market for environmental monitoring is dominated by global platforms (like IQAir, AccuWeather, and generic stock weather applications on smartphones) and localized government portals (like the SAMEER app by the Central Pollution Control Board in India).
   - Global platforms excel at offering worldwide coverage and slick user interfaces but generally rely on generalized US-EPA AQI scaling, which doesn't reflect local regulatory standards or localized sub-district microclimates.
   - Government tools often use the correct local mathematical models (like Indian NAQI) but frequently suffer from poor user experience, lack of historical analytical tools, no interactive predictive health dashboards, and an absence of automated personal alerting systems built on top of the data.
   - State-of-the-art systems currently treat weather and air quality as siloed metrics, rarely combining them to produce a unified, actionable health-risk assessment (like combining Humidity, Temperature, and PM2.5 levels into a single "Outdoor Risk Factor").

7. **List out the known ways about how others have tried to solve the same or similar problems? Indicate the disadvantages of these approaches. In addition, please identify any prior art documentation or other material that explains or provides examples of such prior art efforts.**

S. No.	Existing state of art	Drawbacks in existing state of art	Overcome (how your invention is overcoming the drawback)
1.	Global Weather Applications (e.g., Apple Weather, AccuWeather)	Uses US-EPA AQI standards globally; lacks district-level granularity for India; treats weather and AQI	Implements the Indian NAQI (CPCB) formula strictly. Covers 780+ specific Indian districts. Combines AQI with

		as separate, unrelated data points.	Steadman's Heat Index to provide a unified, localized health verdict.
2.	Government Portals (e.g., SAMEER App, CPCB website)	Poor user interface; lacks automated historical analysis tools (charts/trends); no automated user-configurable threshold push/email notifications.	Features a modern, Glassmorphism-based PWA interface with Chart.js analytics. Includes a BackgroundService that automatically checks user-defined thresholds and triggers HTML email alerts without manual app checking.
3.	Hardware-centric IoT Dashboards	Highly localized but requires deploying expensive physical sensor infrastructure. Cannot scale easily to cover an entire nation's districts comprehensively.	Leverages a hybrid software approach: it aggregates data via highly available satellite/API endpoints (OpenWeatherMap) and processes it locally via a centralized .NET 8 Background Engine, achieving nationwide coverage without hardware deployment costs.

8. **List the Technical features and Elements of the invention along with the Description of your invention from start to end.**

The invention is a unified web-based software dashboard (ASP.NET Core 8.0 MVC) comprising the following technical elements:

1. The Telemetry Engine: An IHostedService daemon that runs asynchronously, pinging external data lakes in batched intervals to prevent rate-limiting, and storing raw temporal data in a SQL Server database.
2. The NAQI Calculator: A C# algorithmic module that parses raw pollutant concentrations (µg/m³) and maps them against the Indian CPCB breakpoint matrices to calculate sub-indices and determine the leading pollutant.
3. The Thermal Comfort Module: An algorithmic implementation of Steadman's Heat Index that takes raw temperature and relative humidity arrays to calculate the perceived physiological temperature.
4. The Geospatial UI: A Leaflet.js-powered interactive front-end that dynamically injects JSON district data into a rendered map layer.
5. The Alert Broker: A subscription-based event listener that triggers SMTP protocol emails when stored AQI thresholds for specific users are breached during a background snapshot cycle.

Description from Start to End: When the system boots, the background service begins iterating over 780+ pre-seeded Indian districts. It pulls raw JSON arrays containing meteorological and pollutant data. This data is piped into the NAQI mathematical model and the Heat Index calculator. The processed metrics are saved as relational snapshots in the database. Simultaneously, the system queries a subscription table; if the newly calculated NAQI for District 'X' exceeds User 'Y's configured threshold, an automated alert payload is dispatched. On the client side, a user accessing the PWA is served a customized dashboard. Upon clicking a district on the geospatial map, the application queries the backend, instantly retrieving the latest processed snapshot, generating historical trend charts, and rendering a definitive "Health Verdict."

9. **List out the features of your invention which are believed to be new and distinguish them over the closest technology.**

   Integrated Algorithmic Health Assessment: Unlike generic apps that just show "AQI: 150" and "Temp: 35°C", this invention synthesizes these variables using the Indian NAQI standard and Steadman's formula to compute a holistic "Safety Verdict," making complex environmental data immediately actionable for laypersons.

   Server-Side Automated Snapshotting with Event Triggering: The architecture inherently solves the problem of "pull-based" monitoring. The user doesn't need to open the app. The server acts autonomously, capturing hourly state snapshots for 780+ geographies and pushing alerts based on personalized, granular thresholds.

   Dual-Strategy PWA Architecture for Environmental Data: It employs a specialized Service Worker manifest that caches static UI assets (Cache-first) but gracefully degrades API calls (Network-first), ensuring that users in remote Indian districts with highly unstable internet can still view the most recent cached environmental analysis.

10. **Has the invention been built or tested or implemented? If yes please provide the Efficiency/Efficacy details of the invention**

    Yes. The system has been fully built as a prototype full-stack web application.

    - Efficacy Details: The background engine successfully retrieves, calculates, and stores data for over 780 districts. Testing demonstrates that the NAQI mathematical logic perfectly matches manual CPCB breakpoint calculations. The alerting mechanism successfully dispatches threshold notifications within milliseconds post-calculation. The PWA achieves a near-instant load time on repeat visits due to aggressive Service Worker caching, and the Chart.js visualizers seamlessly render 30-day historical trends pulling from the SQL backend without noticeable latency.

11. **Briefly state when and how you first conceived this idea?**

    The idea was conceived during a period of severe air quality drops combined with unexpected heatwaves. I noticed that while I had access to various weather and pollution apps, none of them spoke to the specific Indian regulatory context (NAQI), nor did they tell me if it was actually safe to go outside by combining the heat and pollution metrics. The need for a centralized, automated alerting dashboard that covered the granular district where I lived (rather than just the nearest major metro city) sparked the architectural design for EnviroWatch.

12. **Have you sold, offered for sale, publicly used or published anything related to this invention? If yes, please briefly explain the dates and circumstances. List those individuals to whom you have revealed your invention. Were non-discloser documents signed prior to discloser in each case? Please state any deadlines of which you may be aware for filing an application on this invention.**
    No.

13. **Include any reasons that your invention would not have been obvious to someone of average skill in the art.**
    A developer of average skill would typically rely on pre-packaged API endpoints (e.g., just displaying the US-AQI number provided directly by OpenWeatherMap). It requires a non-obvious synthesis of domains (environmental science, backend daemon engineering, and front-end geospatial mapping) to realize that raw API data must be intercepted, decoupled, and re-processed through localized Indian algorithms (CPCB breakpoints) and combined with human thermal comfort models (Heat Index) before being served to the client layer.

14. **Additional comments by inventor (if you want to give more details out of scope of this IDF).**
    No comment at this time

15. Drawings/Flowchart/Table

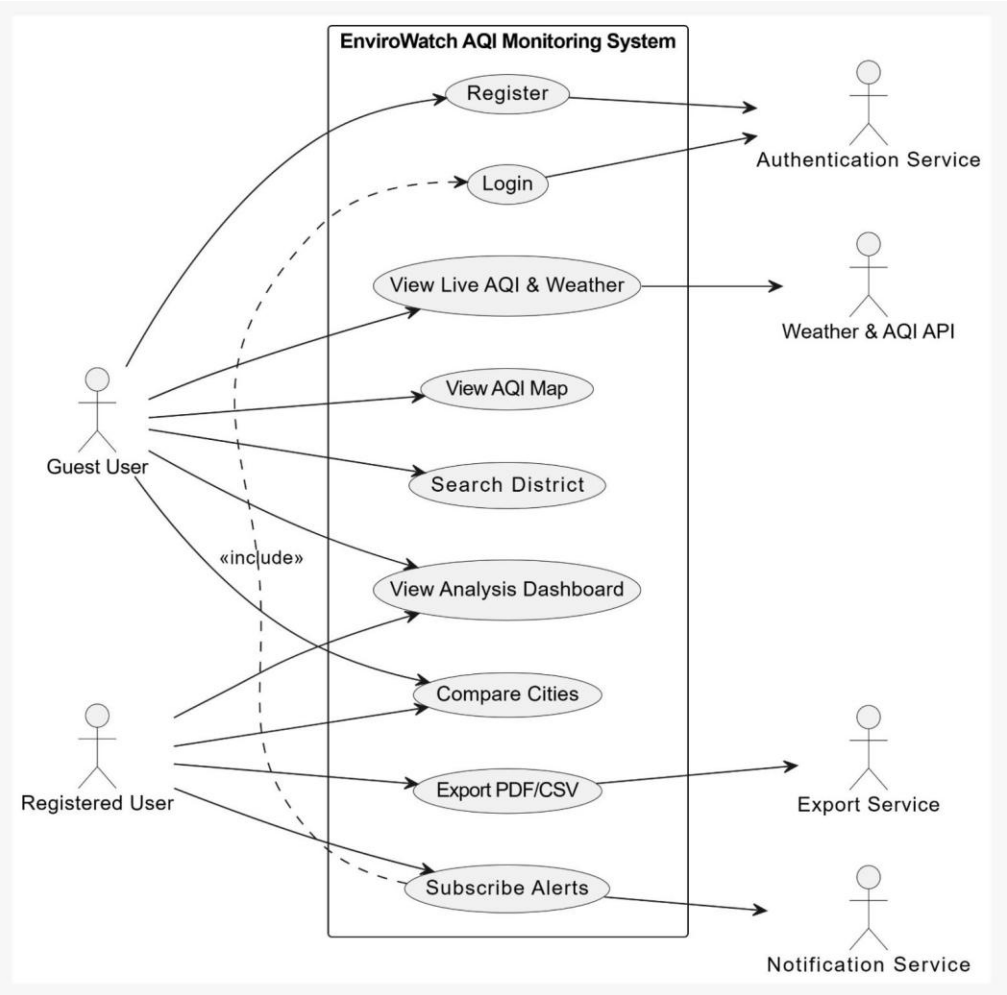


Figure - Use Case Diagram

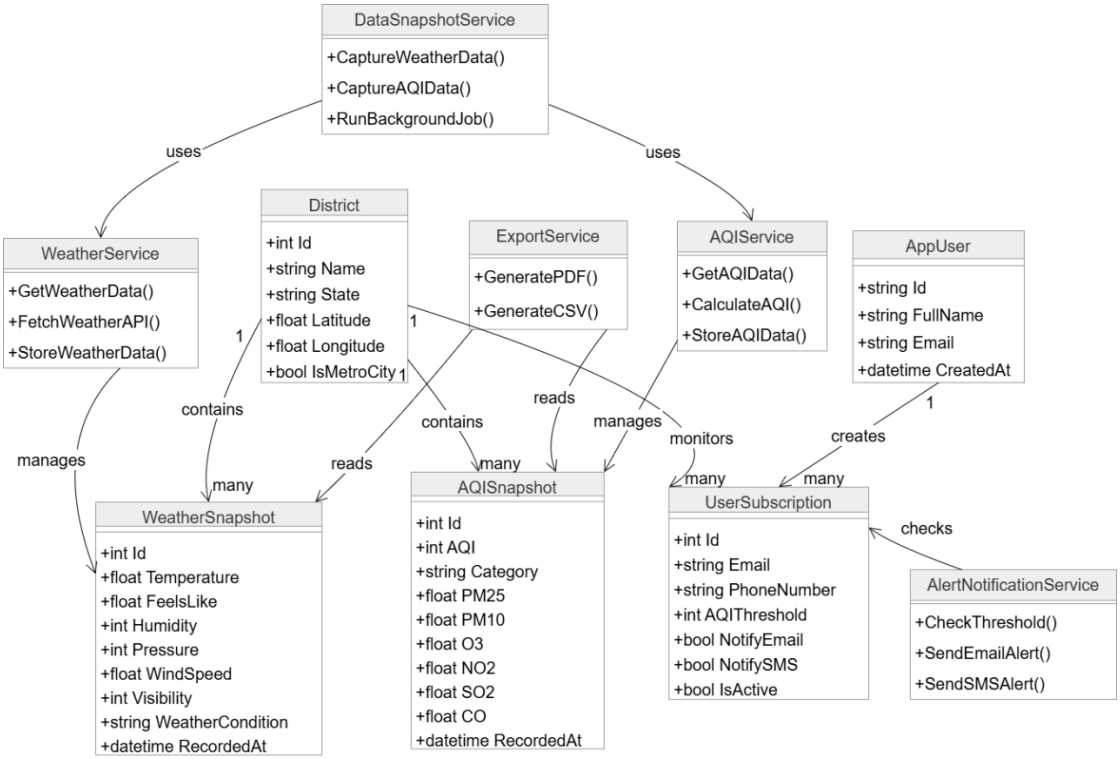


Figure - Class Diagram

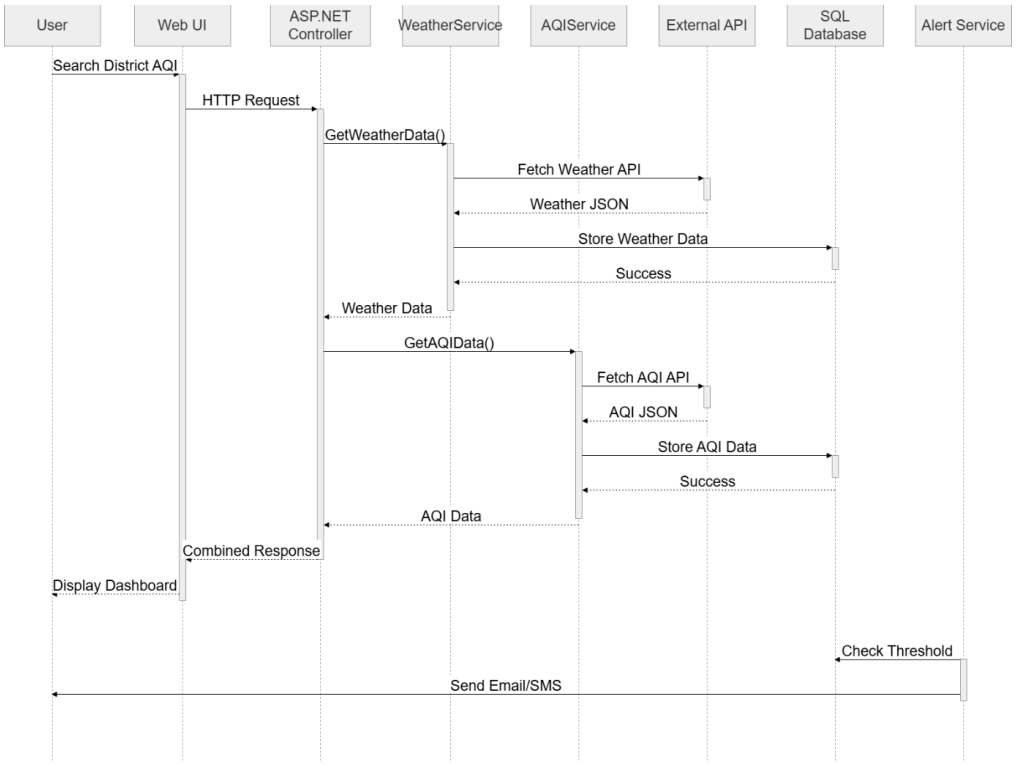


Figure - Sequence Diagram

Contact Details: Aayush Gupta, Email:mu@ennobleip.com, Phn- +91 92891 50390

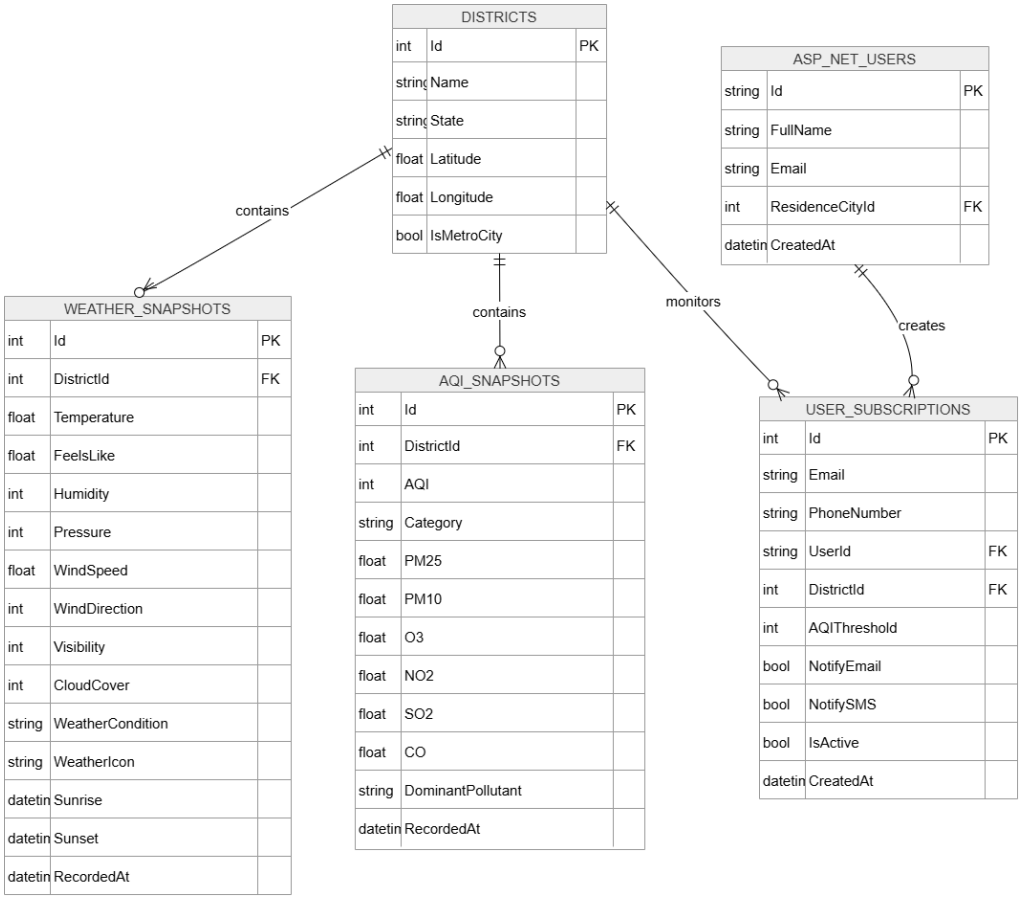


Figure - Physical ER Diagram

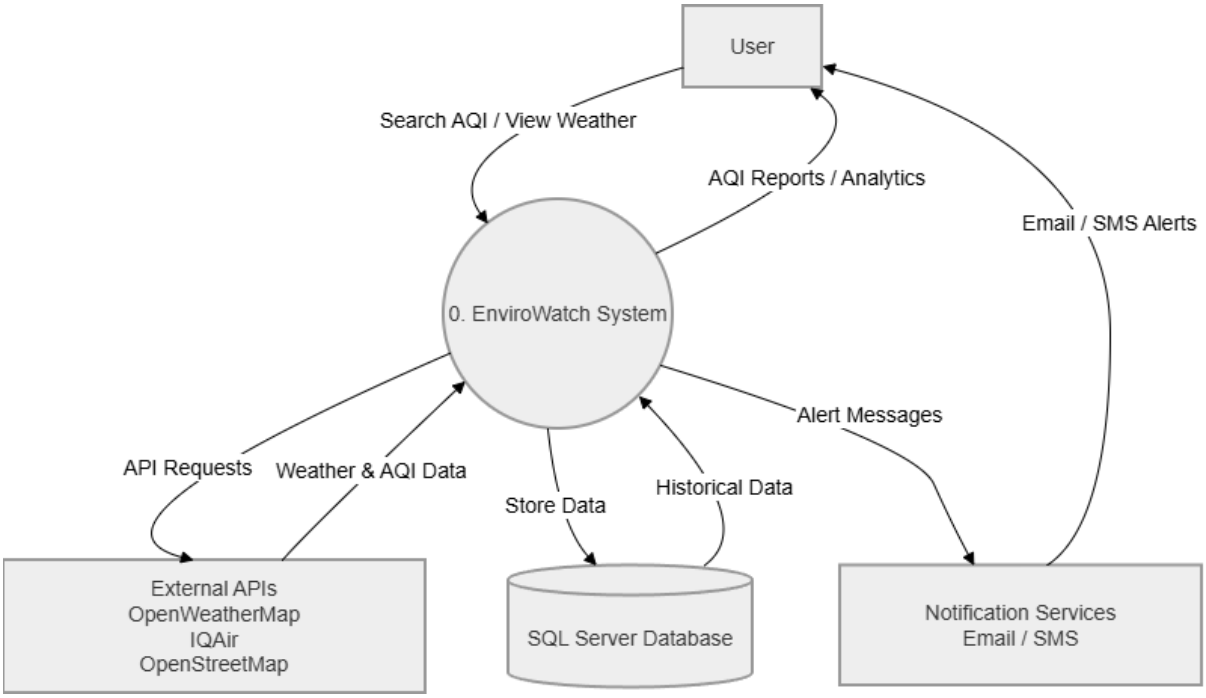
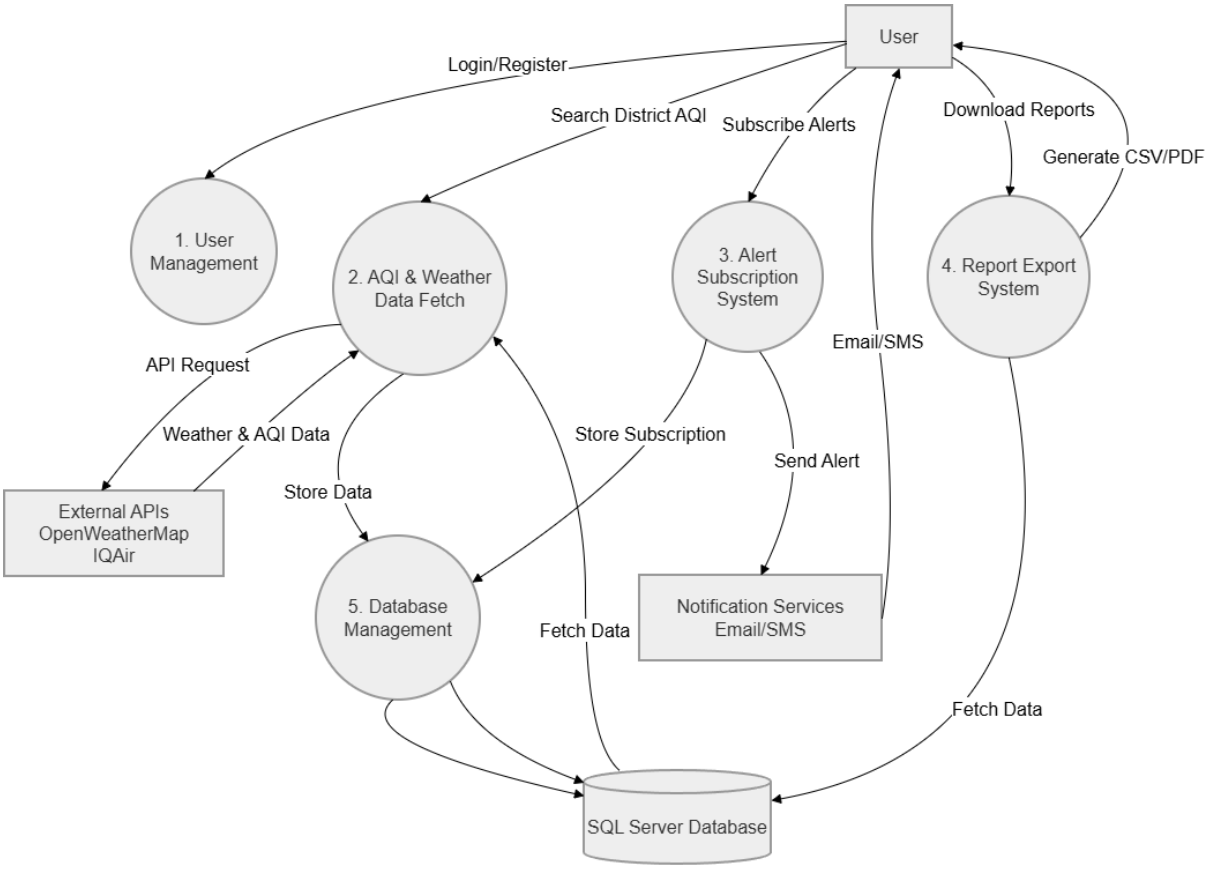


Figure - DFD-0

Contact Details: Aayush Gupta, Email:mu@ennobleip.com, Phn- +91 92891 50390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1. User Management
2. AQI & Weather Data Fetch
3. Alert Subscription System
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Store Data
Store Subscription
Send Alert
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Notification Services Email/SMS
Fetch Data
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Figure - DFD-1